FLOWGUARD: A MULTI-AGENT DIGITAL TWIN SYSTEM FOR INTELLIGENT CROWD MANAGEMENT

Manisha M
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
manisha.m.2024.cse@rajalakshmi.edu.in

Mathu Methaa K
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
Mathumethaa.k.2024.cse@rajalakshmi.edu.in

Abstract

Crowd congestion and poorly coordinated emergency response at public venues remain a recurring safety problem, especially during mass entry, exit, and incidents at large gatherings. Most existing crowd-management systems are either manual, with no simulated decision support, or are commercial digital twins built for a single fixed venue with heavy sensor dependency. This paper presents FlowGuard, a multi-agent digital twin that can be generated for any real-world location selected on a map, rather than one hardcoded venue. Real gate, footpath, road, and parking data is obtained from OpenStreetMap, and venue capacity is estimated by combining geospatial polygon area with classical satellite-image analysis. Four Belief-Desire-Intention (BDI) structured agents (Crowd, Route, Safety, and Emergency) perceive the digital twin's state, while a utility-based Coordinator scores candidate action plans to select the response that minimizes evacuation time, peak density, and safety risk. Dijkstra's and A* algorithms compute real shortest and safest paths, and a Constraint Satisfaction Problem (CSP) solver assigns crowd groups to exits under capacity constraints using backtracking. The system further performs graph-theoretic risk-point detection and routes emergency responders to incidents alongside ranked nearby hospitals. No machine learning is used; every decision is traceable to an explicit algorithm or rule. The system was evaluated on three real stadium twins in Tamil Nadu, India, where blocking a gate or declaring an incident triggered automatic rerouting of simulated crowd agents and dispatch of an emergency responder with a computed route, ETA, and hospital choice.

Keywords— *Digital twin, multi-agent system, crowd management, Dijkstra's algorithm, A* search, constraint satisfaction problem, belief-desire-intention agents, OpenStreetMap, emergency response, explainable AI.*

I. INTRODUCTION

Large public gatherings at stadiums, religious sites, and transport hubs present recurring crowd-safety challenges, particularly during mass entry, mass exit, and emergency evacuation. Manual crowd management relies on human stewards with no predictive or simulated decision support, while large-scale commercial digital-twin deployments are typically built for a single, fixed venue with dense proprietary sensor infrastructure [1], [3]. IoT-based crowd-monitoring frameworks have improved real-time awareness of congestion [2], [10], but most such systems stop at detection and do not provide an explainable, automated reasoning layer that decides how the crowd should be redirected.

This paper presents FlowGuard, a multi-agent digital twin system designed to be generated for any real-world location selected by a user, rather than one hardcoded venue. A selected point on a map is converted into a graph of real gates, footpaths, roads, and exits using OpenStreetMap, and the venue's crowd capacity is estimated using geospatial area combined with classical image analysis. Crowd-management decisions are produced by cooperating agents structured on the Belief-Desire-Intention (BDI) model, coordinated through a utility-based decision function rather than fixed if-else rules, with routing performed using Dijkstra's and A* algorithms and exit assignment solved as a Constraint Satisfaction Problem.

The remainder of this paper is organized as follows. Section II reviews related work in crowd monitoring, digital twins, and evacuation routing. Section III describes the proposed system. Section IV presents the system architecture. Section V details the core algorithms. Section VI reports implementation and results. Section VII concludes with future work.

II. LITERATURE SURVEY

Ten studies published between 2021 and 2025 were reviewed to identify the current state of crowd monitoring, digital-twin crowd control, and evacuation routing. Table I summarizes the methodology of each study, its principal drawback, and how FlowGuard addresses that drawback.

TABLE I
SUMMARY OF RELATED WORK AND IMPROVEMENTS IN FLOWGUARD

Study	Methodology	Drawback	Improvement in FlowGuard
Li et al. [1], 2021	Survey of sensor-level, feature-level, and decision-level data fusion for crowd monitoring.	Conceptual taxonomy only; no routing or decision layer implemented.	Implements a working decision pipeline (agents and Coordinator) rather than a taxonomy.
Noor [2], 2023	Hidden Markov Model classifies CCTV footage into normal and abnormal flow blocks (Hajj/Umrah data, 97.3% accuracy).	Needs dense CCTV coverage; detects abnormal behaviour but proposes no response.	No cameras needed (OSM-based twin); adds agent-driven rerouting after detection.
Li, S. [3], 2022	IoT-based architecture for a single stadium management system.	Design-level only; no AI decision layer; one fixed venue.	Works for any venue and adds the missing AI decision layer.
Mu [4], 2023	1,500+ campus WiFi access points used as passive sensors; 900M+ records analysed.	Device-level tracking raises privacy concerns; observation only, no rerouting.	Uses public map data with no tracking; adds active intervention.
Zhong et al. [5], 2024	Four-layer digital-twin architecture (digital, modeling, functional, application) for airport crowd control.	Airport-specific; generalization to other venues not shown.	Same twin-to-decision layering, but location-independent.
Yasufuku and Takahashi [6], 2024	Agent-based simulation with a crowd-management platform-as-a-service; congestion predicted 10 to 20 minutes ahead.	Authors' own sensitivity analysis shows simple linear models miss crowd-dynamics complexity.	Avoids numeric prediction; the utility-based Coordinator scores explicit candidate plans.
Meng et al. [7], 2022	A* with path-density and obstacle cost for multi-storey ship evacuation.	Ship-specific; single algorithm, no multi-agent coordination.	Density-aware routing inside a full multi-agent system with CSP and risk analysis.
Golubić [8], 2023	Review of Dijkstra, A*, and linear programming for evacuation planning.	Expository; no implementation or real-world data.	Implements and tests the same theory on real venue graphs.
Dinh Thi Hong Huyen et al. [9], 2023	Dijkstra with route-capacity constraints and blocked-route re-planning.	Capacity assumed as an input; no agent coordination or exit assignment.	Estimates capacity itself (OSM and image analysis) and adds CSP exit assignment.
Chen and Zhang [10], 2025	4K cameras, infrared sensors, UWB positioning, and MobileNetV3 edge computing in a smart gymnasium.	Heavy hardware dependency; cannot scale to a new venue without installation.	No physical infrastructure; any venue is available from OpenStreetMap.

Across all ten studies, three recurring gaps emerge: (i) crowd-monitoring systems [1], [2], [4], [10] provide detection with limited or no automated decision-making; (ii) evacuation-routing studies [7]–[9] optimize path-finding in isolation, without capacity estimation or multi-agent coordination; and (iii) venue-specific systems [3], [5], [10] depend on fixed infrastructure or a single domain rather than generalizing across locations. FlowGuard closes all three gaps within a single pipeline: real geographic data in place of physical sensors, classical capacity estimation, BDI-structured agents coordinated through a utility-based decision function, and CSP-based exit assignment, generalized to any real-world location rather than one fixed venue.

III. PROPOSED SYSTEM

FlowGuard is designed to automatically construct a crowd-management digital twin for any real-world location selected by the user, instead of a single pre-modeled venue. A registered user selects a location by searching an address or clicking on a map. A geocoding service converts the selection into coordinates, which are sent to the OpenStreetMap Overpass API to retrieve the place polygon, entrances, footpaths, roads, parking areas, and emergency exits within the surrounding area. Where OSM data is incomplete, the user may manually mark gates and exits.

This data is assembled into a weighted graph with real-world distances, and the location's crowd capacity is estimated by combining polygon area with classical HSV-based image segmentation of the corresponding satellite imagery, producing a comfortable/dense/danger capacity range. Four BDI-structured agents then continuously evaluate the twin: the Crowd Agent tracks occupancy against capacity and predicts approaching overload from load-increase rate; the Route Agent computes shortest and safest paths using Dijkstra's and A* algorithms; the Safety

Agent manages active incidents through a priority queue ordered by severity; and the Emergency Agent computes responder routes and ranks nearby hospitals when an incident is declared. A utility-based Coordinator scores every candidate action plan on evacuation time, peak density, exit-queue length, and safety priority, selecting the highest-scoring plan rather than a single fixed rule, with all candidate scores logged for explainability.

IV. SYSTEM ARCHITECTURE

The system architecture follows a layered pipeline, shown in Fig. 1.

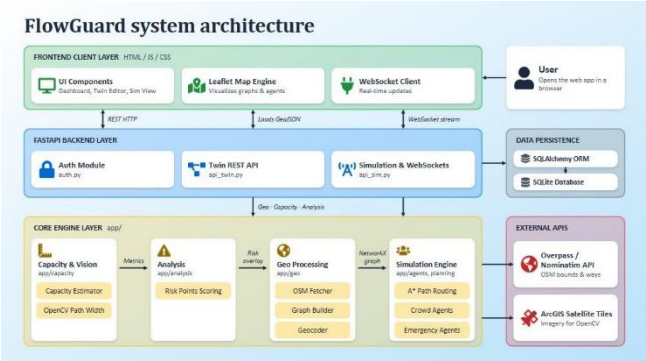


Fig. 1. FlowGuard system architecture.

The backend authenticates each request, builds the real graph for the selected location, and runs the Multi-Agent Layer on every simulation tick. The Coordinator's selected plan is pushed to the visualization layer, which renders live crowd movement, rerouting, and risk layers on the map, while all twins, users, and scenario results are persisted in the database.

V. METHODOLOGY / CORE ALGORITHMS

A. Route Agent — Dijkstra and A Search*

The Route Agent maintains the twin graph G = (V, E) with real distance weights. On a blocked or overloaded node, affected edges are assigned an infinite weight, consistent with evacuation-routing practice in [7]–[9], and Dijkstra's algorithm is re-run from the crowd's current zone to compute the shortest unblocked path to each gate. A* search with a haversine-distance heuristic is used for responder routing on the road network, as this heuristic is admissible for real geographic coordinates and accelerates the search compared to plain Dijkstra.

B. Utility-Based Coordinator

Rather than a fixed rule set, the Coordinator evaluates each candidate action plan p using a weighted utility function: $U(p) = w1\cdot(-T_evac) + w2\cdot(-D_peak) + w3\cdot(-Q_exit) + w4\cdot(S_bonus)$, where T_evac is projected evacuation time, D_peak is projected peak density, Q_exit is the longest resulting exit queue, and S_bonus is a large positive term applied only when an active Safety Agent directive is satisfied, ensuring emergency response always dominates routine routing. The plan with the highest U(p) is selected, and all candidates' scores are logged for explainability.

C. Constraint Satisfaction for Exit Assignment

Crowd groups are assigned to exits as a CSP: variables are groups, domains are reachable unblocked exits, and constraints enforce exit throughput capacity, maximum walking distance, and that each group is assigned to exactly one exit. The solver uses backtracking search with forward checking and a minimum-remaining-values heuristic; if no assignment is feasible, the distance constraint is relaxed incrementally, with each relaxation logged.

D. Risk-Point Detection

Structural risk points are identified using graph-theoretic measures on the twin's graph: articulation points and bridges (single points of failure), betweenness centrality (disproportionate crowd-flow load), and OSM-tagged narrow paths and stairs. Each point receives a weighted risk score used as an additional edge-weight penalty for the Route Agent, so routing naturally avoids high-risk paths when a safe alternative exists.

E. Emergency Agent: Responder Routing and Hospital Ranking

When an incident is declared, the Emergency Agent retrieves hospitals, police stations, and fire stations from OpenStreetMap within a configurable radius of the twin and computes a responder route into the incident using A* search on the road network, with a final footpath segment to the incident point. Reachable hospitals are ranked by a utility that combines travel time, distance, and hospital load. Because no public live feed of hospital occupancy exists, occupancy is a manually set value and is labelled as simulated wherever it is displayed.

VI. IMPLEMENTATION AND RESULTS

The system is implemented in Python using FastAPI, SQLite, and NetworkX for graph construction and pathfinding, with Leaflet for the real-time map frontend. OpenStreetMap's Overpass API and Nominatim geocoding supply real venue data, and OpenCV performs the satellite-image analysis used in capacity estimation. All agent logic is implemented using classical algorithms only, with no machine-learning or deep-learning components.

The system was evaluated on three real Tamil Nadu venues: MA Chidambaram Stadium (Chepauk, Chennai), Rajarathinam Stadium, and Jawaharlal Nehru International Stadium. Twins ranged from 506 nodes / 516 edges (Chepauk) to 1,085 nodes / 1,135 edges (Jawaharlal Nehru

International Stadium), with estimated capacities of 17,807, 40,880, and 48,210 people respectively for Chepauk, Rajarathinam Stadium, and Jawaharlal Nehru International Stadium. Risk analysis was run at full graph resolution and flagged several hundred structural risk points per venue (305 for the Chepauk twin alone, based on narrow paths, junction density, and bottleneck structure), of which a smaller high-severity subset (12 points) is surfaced on the dashboard for operator review.

Under blocked-gate scenarios, the Route Agent recomputed alternate A* paths and the simulated crowd agents were redirected to the remaining open gates, with each decision written step by step to the agent log. When an incident was declared, the Emergency Agent computed a road-network route to the nearest hospital; in one run an ambulance was dispatched to the nearest hospital with an ETA of 0.6 minutes and a simulated hospital load of 60%, while structural risk analysis ran alongside incident handling. A systematic quantitative comparison of evacuation time against an unmanaged nearest-gate baseline over repeated runs is planned as future work.

VII. CONCLUSION AND FUTURE WORK

This paper presented FlowGuard, a location-independent multi-agent digital twin for crowd management that combines real OpenStreetMap data, classical image-analysis-based capacity estimation, BDI-structured agents, a utility-based Coordinator, Dijkstra/A* routing, and CSP-based exit assignment to produce fully explainable crowd-management decisions without machine learning. Testing across three real stadium venues showed that a single agent pipeline adapts to different real graphs and capacities without venue-specific tuning, and that blocked-gate and incident scenarios are handled through automatic rerouting of simulated crowd agents and dispatch of an emergency responder with a computed route, ETA, and hospital choice.

Future work includes a systematic quantitative comparison of evacuation time and peak density against an unmanaged nearest-gate baseline over repeated runs; a transparent, non-ML demand-forecasting module driven by holiday, festival, and weather data; live road-traffic integration for responder routing; real visitor-facing notifications; and validation against publicly reported real-world crowd-congestion incidents.

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